Data Preparation/Feature Engineering

1. Overview

Provide a brief overview of the data preparation and feature engineering phase, outlining its significance in the machine learning project.

2. Data Collection

Describe the source(s) of the dataset used in the project and any preprocessing steps taken during data collection.

3. Data Cleaning

Outline the steps taken to clean the raw data. Include details on handling missing values, outliers, and any other data quality issues.

4. Exploratory Data Analysis (EDA)

Summarize the exploratory data analysis performed on the dataset. Include visualizations and key insights gained during this phase.

5. Feature Engineering

Detail the process of creating new features or transforming existing ones. Explain the rationale behind each feature engineering decision.

6. Data Transformation

Describe any data scaling, normalization, or encoding performed on the features. Include code snippets if applicable.

Model Exploration

1. Model Selection

Explain the rationale behind selecting a particular machine learning model for the project. Discuss the strengths and weaknesses of the chosen model.

2. Model Training

Provide details on how the model was trained, including the hyperparameters used and any cross-validation techniques applied.

3. Model Evaluation

Present the evaluation metrics used to assess the model's performance. Include confusion matrices, ROC curves, or any other relevant visualizations.

4. Code Implementation

Include relevant code snippets for both data preparation/feature engineering and model exploration. Use comments to explain key sections of the code.

Please provide images from your data and models (I want to see different visualizations in EDA part as we did pair coding session)!!!!